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Thermodynamic optimizations of the Nd-Sn and Sn-Tb systems

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ABSTRACT

The thermodynamic optimizations of the Nd-Sn and Sn-Tb binary systems were carried out by means of the Calculation of Phase Diagram (CALPHAD) method on the basis of the available experimental data including the thermodynamic properties and phase equilibria. The Gibbs free energies of the liquid, bcc, bct, dhcp and hcp phases were described by the substitutional solution model with the Redlich-Kister equation, while all of the intermetallic compounds (Nd_5Sn_3, Nd_5Sn_4, $Nd_{11}Sn_{10}$, NdSn, Nd_3Sn_5, $NdSn_2$, Nd_3Sn_7, Nd_2Sn_5, $NdSn_3$, Sn_3Tb, βSn_7Tb_3, αSn_7Tb_3, Sn_2Tb, Sn_5Tb_4, $SnTb_4$, $Sn_{10}Tb_{11}$, Sn_4Tb_5 and Sn_3Tb_5) were described by the sublattice model. A set of self-consistent thermodynamic parameters of each phase in the Nd-Sn and Sn-Tb binary systems has been obtained, and the calculated results are in good agreement with the available experimental data.

1. Introduction

In the traditional electronic industry, Pb-Sn solder is widely used in manufacturing and assembly of electronic products owing to its unique combination of material properties and excellent wettability on the Cu substrate. However, the Pb is considered harmful to the human health and environment [1–4]. Therefore, it is important to develop the lead-free solder, which should have the same microstructure stability and mechanical properties as eutectic Pb-Sn solder. So far, a number of investigations have been focused on the lead-free solder alloys such as Sn-Bi, Sn-Ag, Sn-Cu, Sn-In, and Sn-Zn systems [1,5]. Among all these alloys, Sn-Zn system draws much attention because of its excellent properties. For instance, the eutectic temperature (199 °C) of the Sn-Zn alloy is very close to that of eutectic Sn-Pb (183 °C). In addition, the Sn-Zn alloy is less expensive and the mechanical properties such as tensile strength and creep resistance are better than those of the Sn-Pb alloy [6]. Unfortunately, due to the presence of the active zinc, the Sn-Zn based lead-free solders is easily to be oxidized in air, which will result in poor wetting ability [7]. With the addition of the rare earth (RE) elements, the properties such as microstructure, mechanical and wettability of Sn-Zn system were all improved [8,9]. Thus RE elements can be used as additive elements in Sn-Zn-based solders. Therefore, to design the appropriate solder compositions in the Sn-Zn-RE systems effectively, it is important to have a fundamental knowledge about the Sn-RE systems, such as information about the phase equilibria and the related thermodynamic data.

Recently, the Nd-Sn and Sn-Tb binary systems were thermodynamically optimized by Kim et al. [10,11] with using the Modified Quasichemical Model for liquid phase. However, with the CALPHAD approach [12], the Gibbs free energies of solution phases are described by the substitutional solution model and those of intermetallic compounds are modeled by the sublattice model, which make the thermodynamic parameters like the Gibbs energy of mixing of solution and the Gibbs energy of formation of the compounds are different from the work of Kim et al. [10,11]. Considering the compatibility of the thermodynamic parameters, the thermodynamic optimizations of the Nd-Sn and Sn-Tb binary systems have been carried out by using the CALPHAD method, which was adopted in the process of the development of the lead-free thermodynamic database in our group. The thermodynamic optimizations of the Nd-Sn and Sn-Tb binary systems will provide important thermodynamic information for the multicomponent Sn-Zn-RE alloy systems, and the present study is part of the thermodynamic database development of the lead-free solder alloys.

2. Thermodynamic

2.1. Solution phases

In the Sn-RE (RE = Nd, Tb) binary systems, the Gibbs free energies of the liquid, bcc, bct, dhcp, and hcp phases are described by the

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substitutional solution model, as follows:

$$G_m^{\Phi} = \sum_{i=Sn,RE} {}^0G_i^{\Phi} x_i + RT \sum_{i=Sn,RE} x_i ln x_i + \Delta^E G_m^{\Phi} \quad (1)$$

where ${}^0G_i^{\Phi}$ is the Gibbs free energy of the pure component i in the respective reference state with Φ phase, which is taken from the SGTE pure element database [13]; RE means the element Nd or Tb; X_i denotes the mole fraction of the component i; R is the gas constant; T is the absolute temperature; and the term $\Delta^E G_m^{\Phi}$ presents the excess Gibbs free energy, which is expressed by a Redlich-Kister polynomial [14] as:

$$\Delta^E G_m^{\Phi} = x_{Sn} x_{RE} \sum_{m=0}^{n} {}^mL\Phi_{Sn,RE}(x_{Sn} - x_{RE})^m \quad (2)$$

$${}^mL_{Sn,RE}^{\Phi} = a + bT \quad (3)$$

where ${}^mL_{Sn,RE}^{\Phi}$ is the binary interaction parameter, and the coefficients of *a and b* are constants to be evaluated on the basis of available experimental data.

2.2. Stoichiometric intermetallic compounds

The intermetallic compounds of the Nd_5Sn_3, Nd_5Sn_4, $Nd_{11}Sn_{10}$, NdSn, Nd_3Sn_5, $NdSn_2$, Nd_3Sn_7, Nd_2Sn_5, and $NdSn_3$ phases in the Nd-Sn binary system and the Sn_3Tb, βSn_7Tb_3, αSn_7Tb_3, Sn_2Tb, Sn_5Tb_4, $SnTb_4$, $Sn_{10}Tb_{11}$, Sn_4Tb_5, and Sn_3Tb_5 phases in the Sn-Tb system are all treated as stoichiometric phases and described by the sublattice model [15]. The Gibbs energy per mole of formula unit Sn_pRE_q can be expressed, as follows:

$${}^0G_{Sn:RE}^{Sn_pRE_q} = p^0G_{Sn}^{SER} + q^0G_{RE}^{SER} + \Delta^0 G_f^{Sn_pRE_q} \quad (4)$$

where the $\Delta^0 G_f^{Sn_pRE_q}$ indicates the standard Gibbs free energy of formation of the stoichiometric compound from pure elements, which is described as:

$$\Delta^0 G_f^{Sn_pRE_q} = a' + b'T \quad (5)$$

where the parameters of a' and b' are evaluated in the present work.

3. Thermodynamic optimizations

The optimization of the thermodynamic parameters was carried out by using the PARROT program in the Thermo-Calc software [16], which can handle various kinds of experimental data. The experimental data of the phase diagram and thermodynamic properties were used as input to the program. Each piece of selected information was given a certain weight based on the importance of data, and changed by trial and error during the assessment, until most of the selected experimental information was reproduced within the expected uncertainty limits.

3.1. The Nd-Sn system

3.1.1. Literature information

The Nd-Sn system consists of four solid solution phases (bcc (βNd), bct (αSn), dhcp (αNd) and hcp (βSn) phases) and nine intermetallic compounds (Nd_5Sn_3, Nd_5Sn_4, $Nd_{11}Sn_{10}$, NdSn, Nd_3Sn_5, $NdSn_2$, Nd_3Sn_7, Nd_2Sn_5 and $NdSn_3$ phases).

Firstly, the melting temperature of $NdSn_3$ was determined to be 1138 °C by Palenzona [17] using dynamic differential calorimetry. Later, with the thermal analysis and microscopic examination, Kulagina et al. [18] studied the Sn-rich region of the phase diagram 75–100 at% Sn) and found the melting temperature of $NdSn_3$ compound phase to be 1137 ± 4 °C, which agreed with the Ref. [17]. Based on the DTA, XRD and metallographic analysis, Eremenko et al. [19] investigated the Nd-Sn system in the composition range of 0–50 at% Sn, where there are two intermetallic compounds: Nd_5Sn_3 (m.p. 1640 ± 10 °C) and Nd_5Sn_4 (m.p. 1500 °C), an eutectic reaction: L (12.0 at% Sn) ↔ Nd_5Sn_3 + βNd at 870 ± 10 °C and an eutectoid reaction: βNd (6 at% Sn) ↔ αNd + Nd_5Sn_3 at 820 °C. With the same method, Eremenko et al. [20] investigated the Nd-Sn system in the composition range of 45–100 at% Sn, and confirmed the melting temperature of Nd_5Sn_3 and Nd_5Sn_4 to be 1660 °C and 1558 ± 11 °C respectively. Four intermediate phases were additionally found: $Nd_{11}Sn_{10}$ (1396 ± 15 °C), NdSn (1257 ± 4 °C), Nd_3Sn_5 (1130 ± 2 °C) and $NdSn_3$ (1170 °C) [20]. However, the investigation was not adequate. The phases of $NdSn_2$, Nd_3Sn_7 and Nd_2Sn_5 were found in the later studies, the eutectoid temperature of Nd_3Sn_5 (1130 ± 2 °C) and the congruent temperature of $NdSn_3$ (1170 °C) were refined to be 1140 °C and 1150 °C respectively.

According to the reported experimental data obtained by Eremenko et al. [19,20], Massalski [21] established the complete Nd-Sn phase diagram, where there are six intermetallic compounds: Nd_5Sn_3 (m.p. 1660 °C), Nd_5Sn_4, $Nd_{11}Sn_{10}$, NdSn, Nd_3Sn_5 and $NdSn_3$ (m.p. 1170 °C), two eutectic reactions: L ↔ NdSn + $NdSn_3$ at 1159 °C and L ↔ βNd + Nd_5Sn_3, and an eutectoidal transformation is also reported: βNd ↔ αNd + Nd_5Sn_3 at 820 °C. Weitzer et al. [22] discovered three additional compounds: $NdSn_2$, Nd_3Sn_7, and Nd_2Sn_5, and clarified the phase relationship between them. Thus, the eutectic reaction between $NdSn_3$ and NdSn reported by Eremenko et al. [20] was unfeasible. Saccone et al. [23] studied the Nd-Sn phase diagram in the 55–80 at% Sn range by differential thermal analysis, X-ray powder diffraction, optical and scanning electron microscopy and electron probe microanalysis. The sample alloys were prepared in an induction furnace and fused in tantalum crucibles with the starting materials, 99.9% Nd and 99.999% Sn. In this range, five intermediate phases were identified: $NdSn_3$ (m.p. 1150 °C), Nd_2Sn_5, Nd_3Sn_7, $NdSn_2$ (m.p. 1180 °C) and Nd_3Sn_5, where the melting temperature of $NdSn_3$ was altered to be 1150 °C from 1170 °C determined by Eremenko et al. [20]. Two eutectic equilibria were found at 1130 °C and 60.5 at% Sn and 1130 °C and 73 at% Sn. On the basis of the reported experimental data [19–22], Saccone et al. [23] modified the Nd-Sn phase diagram, where there are four eutectic reactions: L (73.0 at% Sn) ↔ Nd_2Sn_5 + $NdSn_3$ at 1130 °C, L (60.5 at% Sn) ↔ NdSn + Nd_3Sn_5 at 1130 °C, L (12.0 at% Sn) ↔ Nd_5Sn_3 + (βNd) at 870 °C, L ↔ $NdSn_3$ + (βSn) at 230 °C, six peritectic reactions: L+ Nd_5Sn_3↔ Nd_5Sn_4 at 1558 °C, L + Nd_5Sn_4 ↔ $Nd_{11}Sn_{10}$ at 1396 °C, L+ $Nd_{11}Sn_{10}$ ↔ NdSn at 1257 °C, L+ $NdSn_2$ ↔ Nd_3Sn_7 at 1145 °C, L+ $NdSn_2$ ↔ Nd_3Sn_5 at 1140 °C, L+ Nd_3Sn_7 ↔ Nd_2Sn_5 at 1135 °C and an eutectoid reaction: βNd ↔ αNd + Nd_5Sn_3.

In addition, the enthalpy of formation of the compound $NdSn_3$ phase was carried out by Palenzona [17] using dynamic differential calorimetry with a conventional DTA apparatus. And by means of the direct synthesis calorimetric, Meschel and Kleppa [24] measured the enthalpy of formation of the Nd_5Sn_3 compound. With the Miedema model, Takeuchi and Inoue [25] summarized that the value of the enthalpy of mixing of liquid phase with the reference states of pure liquid metals at 50 at% Sn in the Nd-Sn system is −53 kJ/mol. However, no temperature dependence was given in the work of Ref. [25]. By using liquid-metal solution calorimeter, Peluso and Pool [26] and Lundin and Pool [27] measured the heats of solution of solid (dhcp) Nd in liquid Sn at 750 K in the Nd-dilute composition range. The partial enthalpy of mixing of Nd at infinite dilution at 750 K was calculated to be −251 kJ/mol [26] and −294 kJ/mol [27], respectively. With emf measurements, Kulagina and Bayanov [28] determined the activities of Nd in liquid solution at 773 K, 873 K and 973 K. Later, Matigorova and Bayanov [29] reported the activity values for the same temperature. Based on the reported partial Gibbs energy of Nd, Kober et al. [30] determined the activity of Nd in the two phases (L + $NdSn_3$) at 1000 K.

3.1.2. Thermodynamic optimizations

The optimized phase diagram of 55–80 at% Sn region was constructed based on the work of Saccone et al. [23], and the other composition range based on the experimental data of Eremenko et al. [17–20]. The calculated Nd-Sn phase diagram compared with the experimental data [17–20,23] is shown in Figs. 1, and 2 presents the

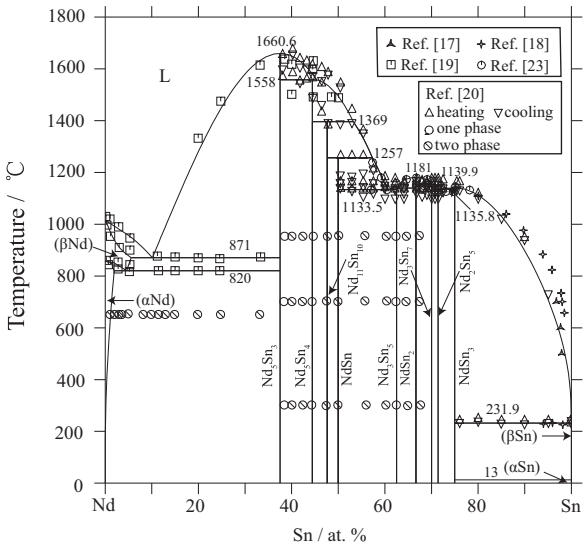

Fig. 1. Calculated phase diagram of the Nd-Sn system compared with the experimental data [17–20,23].

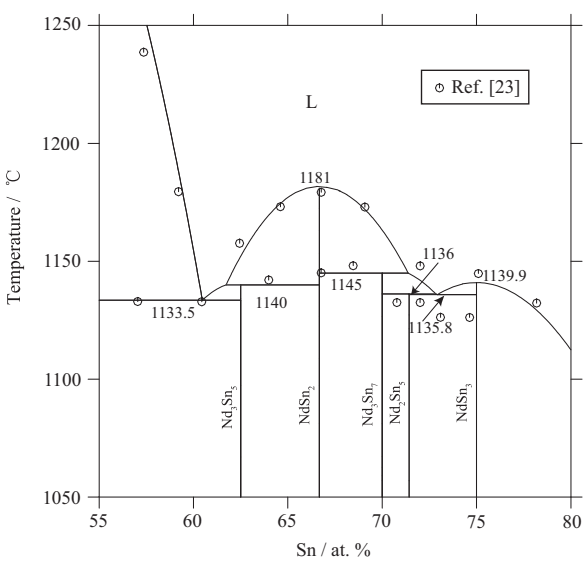

Fig. 2. The enlarged diagram for 55–80 at% Sn range with available experimental data [23].

enlarged diagram of 55–80 at% Sn range with the experimental data [23]. It is seen that the calculated results are in reasonably agreement with the experimental data, except that the temperature (1135.8 °C) of the eutectic reaction (L ↔ Nd_2Sn_5 + $NdSn_3$) is slightly higher than the experimental data (1130 °C) [23], and the melting temperature of the $NdSn_3$ (1139.9 °C) is slightly lower than the experimental data (1150 °C). The discrepancy between the present evaluations and the experimental data is that there are three peritectic reactions and two eutectic reactions in the small range, and the liquidus is more complex. During the process of the optimization, the maximum solubility of Sn in bcc-Nd at eutectic temperature and the eutectoid composition were considered to adjust to fit the experimental data, which made the liquidus around 90 at% Sn in the optimized phase diagram slightly lower than the experimental data, and also slightly lower than the work of Kim [10]. Whereas, the liquidus is close to the work of Eremenko [20]. Based on the Kulagina's work [18], phase diagram of Nd-Sn system was re-investigated by Eremenko et al. [20], therefore, the data in Eremenko's work [20] was considered to be more reasonable than the

Table 1
Thermodynamic parameters in the Nd-Sn binary system optimized in the present work.

Parameters in each phase (J/mol)
Liquid phase, format (Nd, Sn)
${}^{0}L^{Liq}_{Nd,Sn} = -235{,}000 + 30.000\,T$
${}^{1}L^{Liq}_{Nd,Sn} = 8000 + 9.028\,T$
${}^{2}L^{Liq}_{Nd,Sn} = 27{,}482 + 6.200\,T$
${}^{3}L^{Liq}_{Nd,Sn} = 8200–3.000\,T$
Bcc phase, format (Nd, Sn)
${}^{0}L^{Bcc}_{Nd,Sn} = -235{,}000 + 26.300\,T$
${}^{1}L^{Bcc}_{Nd,Sn} = -1000 + 15.000\,T$
${}^{2}L^{Bcc}_{Nd,Sn} = 56{,}500–5.000\,T$
Dhcp phase, format (Nd, Sn)
${}^{0}G^{Dhcp}_{Sn} = {}^{0}G^{Bct}_{Sn} + 5000$
${}^{0}L^{Dhcp}_{Nd,Sn} = -170{,}880$
${}^{1}L^{Dhcp}_{Nd,Sn} = 20.000\,T$
Nd_5Sn_3 phase, format $(Nd)_{0.625}\,(Sn)_{0.375}$
$\Delta G^{Nd_5Sn_3}_{Nd:Sn} = -73{,}000 + 6.320\,T$
Nd_5Sn_4 phase, format $(Nd)_{0.556}\,(Sn)_{0.444}$
$\Delta G^{Nd_5Sn_4}_{Nd:Sn} = -71{,}000 + 3.374\,T$
$Nd_{11}Sn_{10}$ phase, format $(Nd)_{0.524}(Sn)_{0.476}$
$\Delta G^{Nd_{11}Sn_{10}}_{Nd:Sn} = -70{,}260 + 2.774\,T$
NdSn phase, format $(Nd)_{0.5}(Sn)_{0.5}$
$\Delta G^{NdSn}_{Nd:Sn} = -69{,}640 + 2.592\,T$
Nd_3Sn_5 phase, format $(Nd)_{0.375}(Sn)_{0.625}$
$\Delta G^{Nd_3Sn_5}_{Nd:Sn} = -66{,}800 + 3.406\,T$
$NdSn_2$ phase, format $(Nd)_{0333}(Sn)_{0.667}$
$\Delta G^{NdSn_2}_{Nd:Sn} = -65{,}850 + 3.865\,T$
Nd_3Sn_7 phase, format $(Nd)_{0.3}(Sn)_{0.7}$
$\Delta G^{Nd_3Sn_7}_{Nd:Sn} = -65{,}400 + 5.570\,T$
Nd_2Sn_5 phase, format $(Nd)_{0.286}(Sn)_{0.714}$
$\Delta G^{Nd_2Sn_5}_{Nd:Sn} = -65{,}150 + 6.340\,T$
$NdSn_3$ phase, format $(Nd)_{0.25}\,(Sn)_{0.75}$
$\Delta G^{NdSn_3}_{Nd:Sn} = -64{,}500 + 8.255\,T$

Kulagina's work [18] to some extent. What's more, the temperature dependence of Nd solubility in liquid Sn was expressed by the equation, log (at% Nd) = 2.088–3826 T^{-1}. The solubilities were smaller than the values from the emf measurements [28]. Thus, the calculated liquidus in Sn-Nd system in this work is acceptable. A set of complete self-consistent thermodynamic parameters describing the Gibbs free energy of each phase in the Nd-Sn system is given in Table 1. The calculated compositions and temperatures for the invariant reactions are compared with the experimental data in Table 2.

With the optimized thermodynamic parameters, the calculated thermodynamic properties are shown in Figs. 3–6. The calculated standard enthalpies of formation of the compounds with reference states of the Nd (dhcp) and Sn (bct) phases compared with the experimental data [17,24] are presented in Fig. 3. Similar to the Kim's work [10], the calculated enthalpy of formation of $NdSn_3$ in the present work is more positive than the experimental data [17], which is reasonable by considering the fact that the variation of the experimental enthalpies of formation of $RESn_3$ (RE = La, Ce, Pr, Nd, Sm, Gd and Yb) determined by Palenzona [17] was too large. All the enthalpies of formation of the compounds were assessed by simultaneously considering the phase equilibrium and thermodynamic properties. The calculated enthalpies of mixing of liquid phase at 2473 K with reference states of pure Nd (liquid) and Sn (liquid) in the Nd-Sn system compared with the predicted data [25] are shown in Fig. 4. Even though no temperature dependence was given, the predicted data [25] was used in many published papers [31–33], which were considered to be accepted to some extent. Therefore, owing to lack of the experimental data of the

Table 2
Comparison between the calculated and experimental results [17,18,20,23] in Nd-Sn system.

Invariant Reaction	Reaction type	Composition at% Sn			Temperature °C	Reference
L ↔ Nd_5Sn_3	Congruent		37.5		1660	[20]
			37.5		1660.6	This work
L+ Nd_5Sn_3↔ Nd_5Sn_4	Peritectic	~49	37.5	44.4	1558 ± 11	[20]
		45.3	37.5	44.4	1558	This work
L+ Nd_5Sn_4 ↔ $Nd_{11}Sn_{10}$	Peritectic	~55	44.4	47.6	1396 ± 15	[20]
		53.2	44.4	47.6	1396	This work
L+ $Nd_{11}Sn_{10}$ ↔ NdSn	Peritectic	~57	47.6	50	1257 ± 4	[20]
		57.3	47.6	50	1257	This work
L ↔ $NdSn_2$	Congruent		66.7		1180	[23]
			66.6		1180.7	This work
L+ $NdSn_2$ ↔ Nd_3Sn_7	Peritectic	~72	66.7	70	1145	[23]
		71.2	66.7	70	1145	This work
L+ $NdSn_2$ ↔ Nd_3Sn_5	Peritectic	~61	66.7	62.5	1140	[23]
		61.7	66.7	62.5	1140	This work
L+ Nd_3Sn_7 ↔ Nd_2Sn_5	Peritectic	~73	70.0	71.4	1135	[23]
		72.8	70.0	71.4	1136.2	This work
L ↔ $NdSn_3$	Congruent		75		1138	[17]
			75		1137	[18]
			75		1170	[20]
					1150	[23]
			75		1139.9	This work
L ↔ Nd_2Sn_5 + $NdSn_3$	Eutectic				1159 ± 4	[20]
		73	71.4	75	1130	[23]
		72.9	71.4	75	1135.8	This work
L ↔ NdSn + Nd_3Sn_5	Eutectic	60.5	50	62.5	1130	[23]
		60.5	50	62.5	1133.5	This work
L ↔ Nd_5Sn_3 + (βNd)	Eutectic	12	37.5	7	870	[20]
		10.1	37.5	5.3	871	This work
(βNd) ↔ Nd_5Sn_3 + (αNd)	Eutectoid	6	37.5	2	820	[20]
		4.6	37.5	1.8	820.1	This work
L ↔ $NdSn_3$ + (βSn)	Eutectic		75.0	100	230	[20]
		99.9	75.0	100	231.9	This work

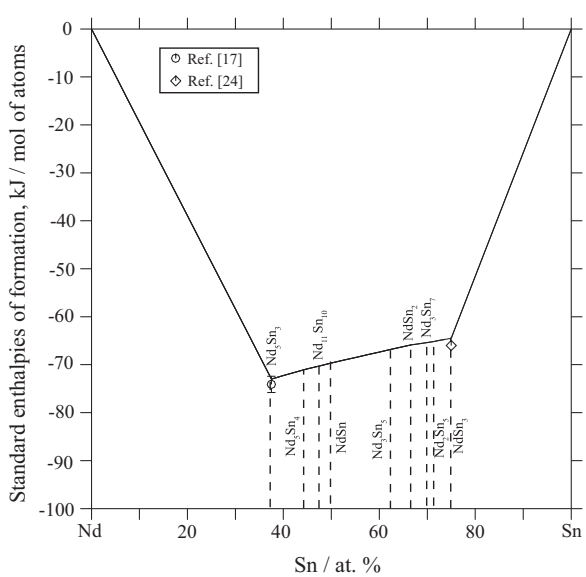

Fig. 3. Calculated standard enthalpies of formation with reference states of Nd (dhcp) and Sn (bct) compared with the experimental data [17,24].

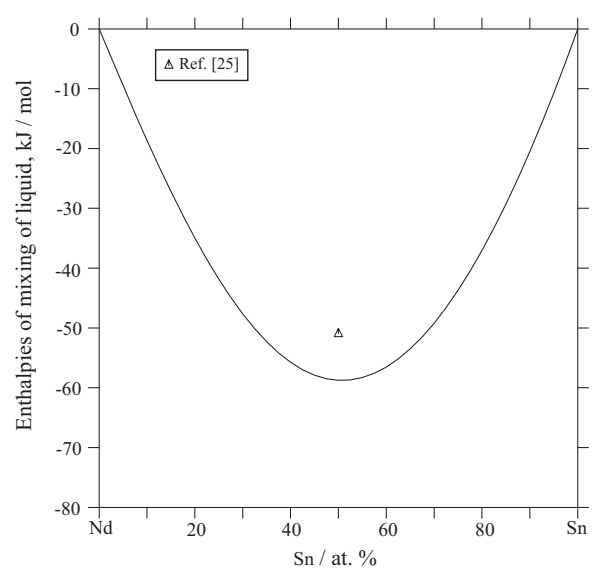

Fig. 4. Calculated enthalpies of mixing of liquid phase at 2473 K with reference states of pure elements of Nd (liquid) and Sn (liquid) compared with the predicted data [25].

enthalpies of mixing of liquid phase, the predicted data [25] was given a certain weight to reproduce the enthalpies of mixing of liquid phase in the present work. It is seen from Fig. 4 that the enthalpies of mixing of liquid is slightly more negative than the predicted data. The calculated integral and partial enthalpies of mixing of the Nd-Sn liquid at 750 K are shown in Fig. 5(a) and (b). Similar to the Kim's work [10], the experimental data of the partial enthalpies of mixing of Nd in the infinitely dilute solution at lower temperature (750 K) [26,27] were not considered in the present work. Calculated activities of Nd at 773–1000 K in liquid Nd-Sn system compared with the experimental data [28–30] are illustrated in Fig. 6(a) and (b), and it can be seen that the calculated activities of Nd at 773–1000 K up to 1 at% Nd are slightly lower than the experimental data [28–30] in Fig. 6(b). However, according to Fig. 6(a), the calculated activities of Nd in Fig. 6(a) are consistent with the experimental data basically.

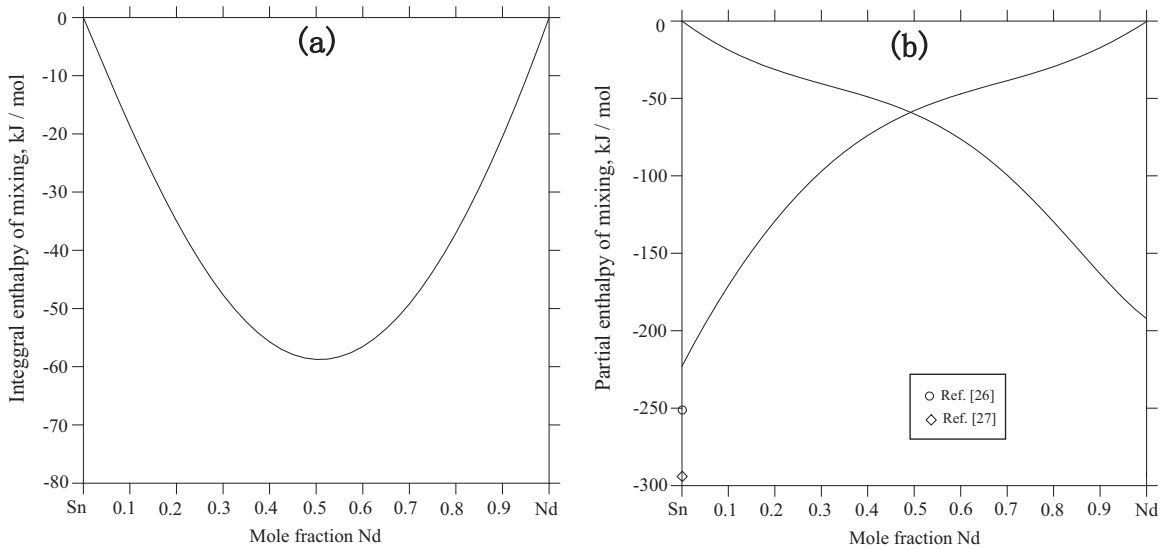

Fig. 5. (a) Integral enthalpy of mixing at 750 K (b) Partial enthalpies of mixing of liquid Sn-Nd at 750 K with experimental data [26,27].

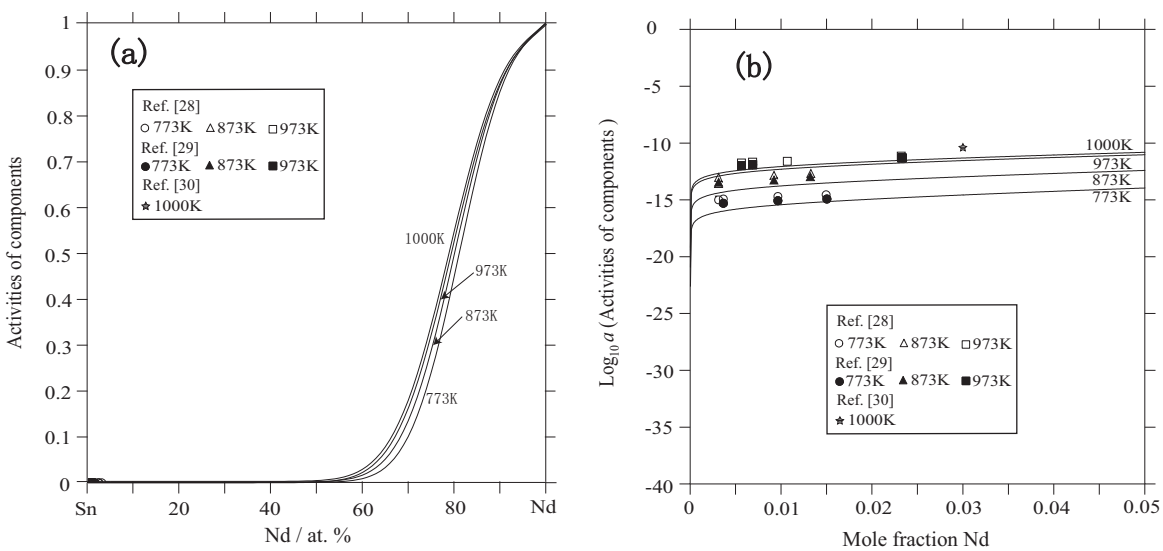

Fig. 6. Calculated activities of components at 773–1000 K in liquid Nd-Sn system compared with the experimental datas [28–30]. (a) linear scale, (b) log scale.

3.2. The Sn-Tb system

3.2.1. Literature information

The Sn-Tb system consists of three solid solution phases (bcc (βTb), bct (αSn) and hcp (βSn and αTb) phases) and nine intermetallic compounds (Sn_3Tb, βSn_7Tb_3, αSn_7Tb_3, Sn_2Tb, Sn_5Tb_4, SnTb, $Sn_{10}Tb_{11}$, Sn_4Tb_5, Sn_3Tb_5 (m.p. 1850 °C)). By using the DTA method, Eremenko et al. [34] firstly investigated the 0.619, 0.64 and 0.908 mol fraction of Tb of the Sn-Tb phase diagram. The melting temperature of Tb_5Sn_3 and the eutectic temperature were determined to be 1850 °C and 1160 °C, respectively. Later, Eremenko et al. [35] determined the Sn-Tb phase diagram over the whole composition range with the DTA, XRD and metallographic methods, and the sample alloys were prepared by arc-melting under a purified argon atmosphere from the initial materials, 99.999% Sn and 99.85% Tb. Seven intermetallic compounds were found: Sn_3Tb_5 (1850 °C, congruent), Sn_4Tb_5 (1710 °C, peritectic), $Sn_{10}Tb_{11}$ (1590 °C, peritectic), SnTb (1259 ± 3 °C, peritectic), Sn_5Tb_4 (1181 ± 6 °C, peritectic), Sn_2Tb (1136 ± 2 °C, peritectic) and Sn_3Tb (852 ± 3 °C, peritectic). There also exist two eutectic reactions: L ↔ Sn_2Tb + Sn_5Tb_4 and L (0.88 at% Tb) ↔ αTb + Sn_3Tb_5 at 1158 ± 8 °C and an allotropy transformation: αSn_3Tb ↔ βSn_3Tb at 762 °C where Eremenko et al. [35] assumed that Sn_3Tb exists in a limited temperature range between 480 °C and 852 °C. Later, according to DTA, microscopic analysis, X-ray diffraction and electron microscopy, Palenzona and Manfrinetti [36] modified the Sn-Tb phase diagram in the 0–40 at% Tb range. In this range, Palenzona and Manfrinetti [36] determined Sn_3Tb (450 °C, peritectic) phase and Sn_2Tb (1150 °C, peritectic) phase and reported that the Sn_3Tb phase was stable down to the room temperature, which are different from the work of Eremenko et al. [35]. And Palenzona and Manfrinetti [36] found the Sn_7Tb_3 (830 °C, peritectic) phase exist in the system with a polymorphic transformation (αSn_7Tb_3 ↔ βSn_7Tb_3 at 745 °C). In the work of Ref. [36], there exists a peritectic reaction: L + Sn_5Tb_4 ↔ Sn_2Tb at 1150 °C, which is considered in the optimization. Based on the Ref. [35,36], Okamoto [37] reviewed and modified the phase diagram of Sn-Tb binary system.

In addition, by using a relaxation method, Szytula et al. [38] measured the heat capacity of Sn_2Tb in low temperature between 1.9 K and 300 K, and the samples of Sn_2Tb were synthesized by arc melting constituent elements of high purity (99.9 for Tb and 99.99% for Sn). The heat capacity of Sn_2Tb showed some anomalies at 13.1 K, 18.7 K and 26.9 K, which are related to the magnetic transformation, the change in the magnetic structure and unknown spurious phase,

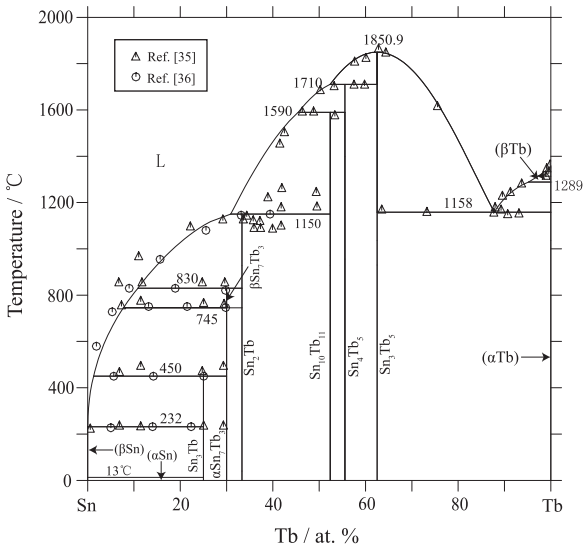

Fig. 7. Calculated phase diagram of the Sn-Tb system compared with the experimental data [35,36].

respectively. The standard entropy of formation was calculated to be 159.15 J·K^{-1} mol^{-1} according to the heat capacity in the work of Szytula [38]. Meschel and Kleppa [39] reported the standard enthalpy of formation of the Sn_3Tb_5 phase at 1473 ± 2 K with the direct synthesis calorimetric methods. Takeuchi and Inoue [25] summarized that the value of the enthalpy of mixing of liquid phase with the reference states of pure liquid metals at 50 at% Tb in the Sn-Tb system is −50 kJ/mol, however, no temperature dependence was given in the work of Takeuchi and Inoue [25].

3.2.2. Thermodynamic optimizations

Fig. 7 illustrates the calculated phase diagram of the Sn-Tb system compared with the experimental data [35,36]. It is apparent that the results agree with the experimental data basically. Owing to its systematic study, the work of Ref. [36] was considered to be more reliable by Okamoto [37]. Therefore, the calculated diagram in the range of 0–40 at% Tb was constructed based on the work of Palenzona and Manfrinetti [36], and the composition above 40 at% Tb based on the work of Eremenko et al. [35]. Since the Sn_5Dy_4 and SnDy phases were found in the Sn-Dy phase diagram of Eremenko et al. [40], but the two phases were not found in the several isothermal phase diagram investigations of the ternary systems containing Dy and Sn [41–43]. These ternary systems were studied by scanning electron microscopy (SEM), electron probe micro-analysis (EPMA) and X-ray analysis. And the samples were arc-melted under purified argon atmosphere with non-consumable tungsten electrode on a water-cooled copper hearth. Considering the similarity with the Sn-Dy system, the Sn_5Tb_4 and SnTb phases in the Sn-Tb system were excluded in the present work. Kim et al. [11] also neglected the Sn_5Tb_4 and SnTb phases in their work based on the Refs. of [44–46]. A set of complete self-consistent thermodynamic parameters describing the Gibbs free energy of each phase in the Sn-Tb system are given in Table 3, and the calculated compositions and temperatures for the invariant reactions compared with the experimental data are listed in Table 4.

Table 3
Thermodynamic parameters in the Sn-Tb binary system optimized in the present work.

Parameters in each phase (J/mol)
Liquid phase, format (Sn, Tb)
${}^0L^{Liq}_{Sn,Tb} = -200{,}200 + 29.573\,T$
${}^1L^{Liq}_{Sn,Tb} = 39{,}600 – 9.500\,T$
${}^2L^{Liq}_{Sn,Tb} = 40000$
${}^3L^{Liq}_{Sn,Tb} = -29{,}300 + 2.100\,T$
Sn_3Tb phase, format $(Sn)_{0.75}(Tb)_{0.25}$
$\Delta G^{Sn_3Tb}_{Sn:Tb} = -43{,}070 + 4.005\,T$
βSn_7Tb_3 phase, format $(Sn)_{0.7}(Tb)_{0.3}$
$\Delta G^{\beta Sn_7Tb_3}_{Sn:Tb} = -50{,}710 + 4.503\,T$
αSn_7Tb_3 phase, format $(Sn)_{0.7}(Tb)_{0.3}$
$\Delta G^{\alpha Sn_7Tb_3}_{Sn:Tb} = -51{,}170 + 4.955\,T$
Sn_2Tb phase, format $(Sn)_{0.667}(Tb)_{0.333}$
$\Delta G^{Sn_2Tb}_{Sn:Tb} = -55{,}780 + 5.504\,T$
$Sn_{10}Tb_{11}$ phase, format $(Sn)_{0.476}(Tb)_{0.524}$
$\Delta G^{Sn_{10}Tb_{11}}_{Sn:Tb} = -70{,}600 + 6.628\,T$
Sn_4Tb_5 phase, format $(Sn)_{0.444}$ $(Tb)_{0.556}$
$\Delta G^{Sn_4Tb_5}_{Sn:Tb} = -71{,}850 + 7.075\,T$
Sn_3Tb_5 phase, format $(Sn)_{0.375}$ $(Tb)_{0.625}$
$\Delta G^{Sn_3Tb_5}_{Sn:Tb} = -73{,}100 + 8.300\,T$

Table 4
Comparison between the calculated and experimental results [34–36] in the Sn-Tb system.

Invariant Reaction	Reaction type	Composition at% Tb			Temperature ℃	Reference
L ↔ Sn_3Tb_5	Congruent		62.5		1850.0	[34]
			62.5		1850.0	[35]
			62.5		1850.0	This work
L + Sn_3Tb_5 ↔ Sn_4Tb_5	Peritectic		62.5	55.6	1710.0	[35]
		52.4	62.5	55.6	1710.0	This work
L + Sn_4Tb_5 ↔ $Sn_{10}Tb_{11}$	Peritectic		55.6	52.4	1590.0	[35]
		45.8	55.6	52.4	1590.0	This work
L ↔ (αTb) + Sn_3Tb_5	Eutectic	89.0	100.0	62.5	1160.0	[34]
		87.2	100.0	62.5	1158 ± 8	[35]
		87.8	100.0	62.5	1158.6	This work
L + Sn_5Tb_4 ↔ Sn_2Tb	Peritectic		44.4	33.3	1150.0	[35]
					1136 ± 2	[36]
		30.9	44.4	33.3	1150.0	This work
L + Sn_2Tb ↔ βSn_7Tb_3	Peritectic		33.3	30.0	830.0	[36]
		11.0	33.3	30.0	830.0	This work
βSn_7Tb_3 ↔ αSn_7Tb_3	Polymorphic				762 ± 4	[35]
			30.0		745.0	[36]
		7.8	30.0		745.0	This work
L + αSn_7Tb_3 ↔ Sn_3Tb	Peritectic		30.0	25.0	450.0	[36]
		1.2	30.0	25.0	450.0	This work
L ↔ (βSn) + Sn_3Tb	Eutectic		100.0	25.0	231.0	[36]
		~0	100.0	25.0	231.8	This work

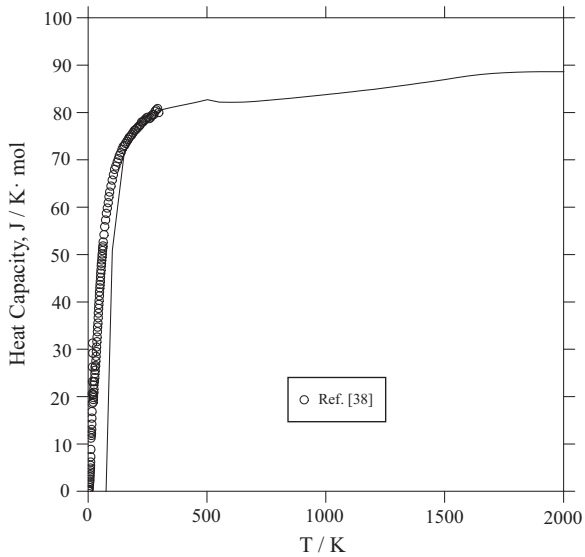

Fig. 8. Heat Capacity of Sn_2Tb with experimental data [38].

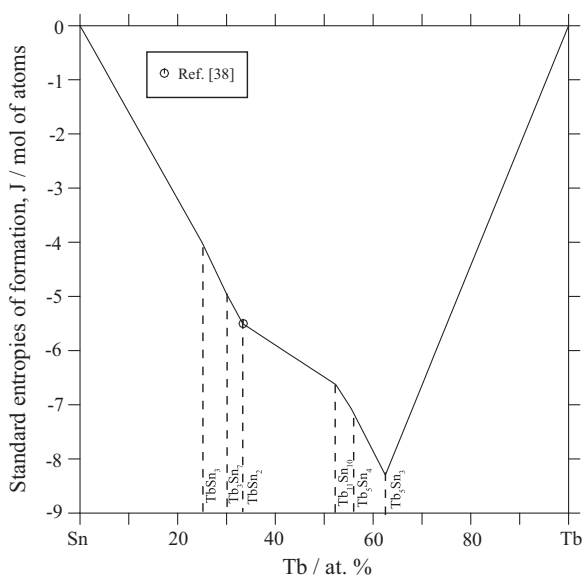

Fig. 9. Calculated standard entropies of formation with reference states of Sn (bct) and Tb (hcp) compared with the experimental data [38].

The calculated thermodynamic properties with the optimized thermodynamic parameters are shown in Figs. 8–11. The calculated heat capacity of Sn_2Tb with experimental data was shown in Fig. 8. The lowest temperature required in the SGTE pure elements database is above 298 K, so the calculated heat capacity below 298 K was not considered. The experimental entropy of Sn_2Tb was reproduced as seen in Fig. 9, the calculated result is consistent with the experimental data [38]. Fig. 10 plots the calculated standard enthalpies of formation with reference states of the Sn (bct) and Tb (hcp) phases in comparison with the experimental data [39]. The calculated standard enthalpies of formation of Tb_5Sn_3 phase are consistent with the experimental data [39]. Fig. 11 shows the calculated enthalpies of mixing of the liquid phase at 2573 K with reference states of pure Sn (liquid) and Tb (liquid) in the Sn-Tb binary system. The calculated value at 50 at% Tb is −50.1 kJ/mol, which is very close to the predicted data of −50.0 kJ/mol [25].

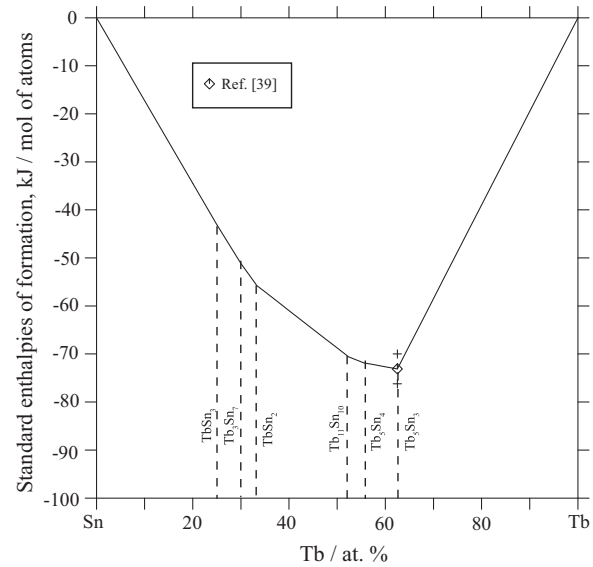

Fig. 10. Calculated standard enthalpies of formation with reference states of Sn (bct) and Tb (hcp) compared with the experimental data [39].

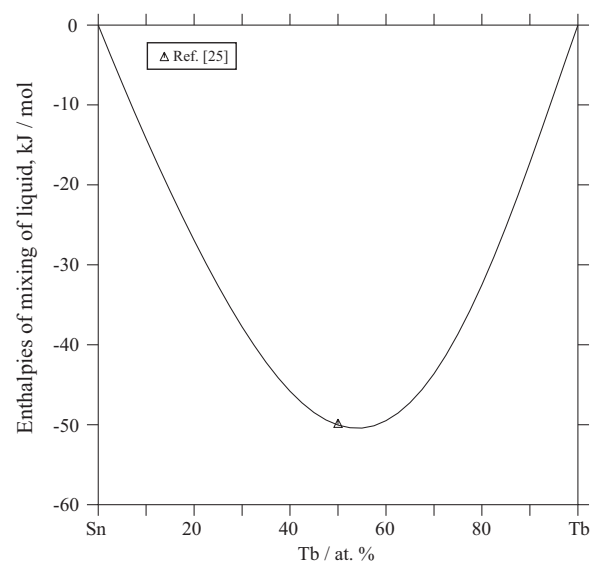

Fig. 11. Calculated enthalpies of mixing of liquid phase at 2573 K with reference states of pure elements of Sn (liquid) and Tb (liquid) compared with the predicted data [25].

4. Conclusions

On the basis of available experimental data, the Sn-RE (RE = Nd, Tb) binary systems have been critically optimized by the CALPHAD method. A consistent set of thermodynamic parameters has been obtained and most experimental information can be satisfactorily reproduced based on the assessed thermodynamic parameters.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.calphad.2017.12.009.

References

[1] C.M.L. Wu, M.L. Huang, J.K. Lai, Y.C. Chan, Developing a lead-free solder alloy Sn-Bi-Ag-Cu by mechanical alloying, J. Electron. Mater. 29 (8) (2000) 1015–1020.
[2] M. Abtew, G. Selvaduray, Lead-free Solders in Microelectronics, Mater. Sci. Eng. Rep. 27 (5–6) (2000) 95–141.
[3] L. Zhang, S.B. Xue, L.L. Gao, G. Zeng, Z. Sheng, Y. Chen, S.L. Yu, Effects of rare earths on properties and microstructure of lead-free solder alloys, J. Electron. Mater. 20 (8) (2009) 685–694.
[4] C.M. Chen, Y.M. Hung, C.P. Lin, W.C. Su, Effect of temperature on microstructural changes of the Sn-9 wt% Zn lead-free solder stripe under current stressing, Mater. Chem. Phys. 115 (1) (2009) 367–370.
[5] C.M.L. Wu, D.Q. Yu, C.M.T. Law, L. Wang, Properties of lead-free solder alloys with rare earth element additions, **Mat. Sci. Eng.** R. 44 (1) (2004) 1–44.
[6] C.M.L. Wu, D.Q. Yu, C.M.T. Law, L. Wang, The properties of Sn-9Zn lead-free solder alloys doped with trace rare earth elements, J. Electron. Mater. 31 (9) (2002) 921–927.
[7] K.L. Lin, T.P. Liu, High temperature oxidation of a Sn–Zn–Al solder, Oxid. Met. 50 (3) (1998) 255–267.
[8] Y.H. Hu, S.B. Xue, H. Wang, H. Ye, Z.X. Xiao, L.L. Gao, Effects of rare earth element Nd on the solderability and microstructure of Sn-Zn lead-free solder, J. Electron. Mater. 22 (5) (2011) 481–487.
[9] D.Q. Yu, J. Zhao, L. Wang, Improvement on the microstructure stability, mechanical and wetting properties of Sn-Ag-Cu lead-free solder with the addition of rare earth elements, J. Alloy. Compd. 376 (1–2) (2004) 170–175.
[10] J. Kim, E. Thibodeau, K. Tetley-Gerard, I. Jung, Critical evaluation and thermodynamic optimization of the Sn–RE systems: part I. Sn–RE system (RE = La, Ce, Pr, Nd and Sm), Calphad 55 (2) (2016) 113–133.
[11] J. Kim, I. Jung, Critical evaluation and thermodynamic optimization of the Sn–RE systems: part II. Sn–RE system (RE = Gd, Tb, Dy, Ho, Er, Tm, Lu and Y), Calphad 55 (2) (2016) 134–156.
[12] L. Kaufman, H. Bernstein, Computer Calculation of Phase Diagrams, Academic Press, New York, 1970.
[13] A.T. Dinsdale, (Version 4.4), SGTE Pure Elem. (Unary) Database (2001) (Version 4.4).
[14] O. Redlich, A.T. Kister, Algebraic representation of thermodynamic properties and the classification of solutions, Ind. Eng. Chem. 40 (2) (1948) 345–348.
[15] M. Hillert, L.I. Stafansson, Regular solution model for stoichiometric phases and ionic melts, Acta Chem. Scand. 24 (10) (1970) 3618–3626.
[16] J.O. Andersson, T. Helander, L. Höglund, P. Shi, B. Sundman, Thermo-calc and DICTRA, computational tools for materials science, Calphad 26 (2) (2002) 273–312.
[17] A. Palenzona, Dynamic differential calorimetry of intermetallic compounds. I. Heat of formation, and heat and entropy of fusion of rare earth-tin com-pounds, Thermochim. Acta 5 (1973) 473–480.
[18] N.G. Kulagina, A.P. Bayanov, N.M. Kulagin, Study of the binary system R-Sn (R = La, Nd, Gd, Er, Lu), Izv. Akad. Nauk SSSR, Met. 3 (1985) 211–216.
[19] V.N. Eremenko, A.M. Khar'kova, I.M. Obushenko, Phase equilibrium diagrams for the neodymium-tin system within a range of 0-50 at% tin, Dop. Akak. Nauk Ukr. RSR, Ser. A 12 (1978) 1132–1136.
[20] V.N. Eremenko, M.V. Bulanova, R.S. Martsenyuk, V.E. Listovinchii, Phase diagram of the neodymium-tin system, Ukr. Khim. Zh. 54 (12) (1988) 1240–1244.
[21] T.B. Massalski (Ed.), Binary Alloy Phase Diagrams, 2nd ed. 3 ASM International, Metals Park, Ohio, 1990.
[22] F. Weitzer, K. Hieblie, P. Rogl, Crystal chemistry and magnetism of neodymium stannides including compounds of the structural series RE_nSn_{3n-2}, J. Solid State Chem. 98 (2) (1992) 291–300.
[23] A. Saccone, D. Maccio, R. Ferro, Phase equilibria of the Nd-Sn system in the 55-80 at % Sn range, J. Alloy. Compd. 201 (1–2) (1993) 9–11.
[24] S.V. Meschel, O.J. Kleppa, Standard enthalpies of formation of some rare earth stannides by high temperature direct synthesis calorimetry, J. Alloy. Compd. 238 (1–2) (1996) 180–186.
[25] A. Takeuchi, A. Inoue, Classification of bulk metallic glasses by atomic size difference, heat of mixing and period of constituent elements and its application to characterization of the main alloying element, Mater. Trans. 46 (12) (2005) 2817–2829.
[26] R.F. Peluso, M.J. Pool, Heats of solution of Pr, Nd, and Sm in liquid Sn at 750 K, in: Proceedings Conference Rare Earth Res., 4th, Phoenix, Ariz., p 269–279, 1965.
[27] C.E. Lundin, M.J. Pool, Heats of mixing in praseodymium-neodymium solid solutions, J. Less-Common Met. 9 (1) (1965) 48–53.
[28] N.G. Kulagina, A.P. Bayanov, Thermodynamic properties of neodymium-tin ($NdSn_3$) and its solutions in molten tin studied by an electromotive forcemethod, Zh. Fiz. Khim. 48 (1974) 466–467.
[29] N.G. Matigorova, A.P. Bayanov, Thermodynamics of the reaction of neody-mium with tin in melts, v sb, Termodin. Svoistva Metal. Splav. (1975) 120–124.
[30] V.I. Kober, I.F. Nichkov, S.P. Raspopin, V.I. Kuz'minykh, Thermodynamic properties of saturated solutions of neodymium with low-melting point metals, in: Proceedings of the Izd. Nauka Kazakhskoi SSR, 2, p 72–76, 1972.
[31] C. Wang, Y. Guo, Y. Lu, et al., Thermodynamic Assessment of the Ti-Ir System, J. Phase Equilib. 35 (3) (2014) 269–275.
[32] X.J. Liu, S.X. Gan, Z.S. Li, et al., Thermodynamic assessment of the Ce–Pt system, Calphad 35 (4) (2011) 551–555.
[33] F. Gao, S.L. Wang, C.P. Wang, et al., Thermodynamic assessments of the Sb–La and Sb–Tb systems, Calphad 37 (37) (2012) 158–162.
[34] V.N. Eremenko, M.V. Bulanova, P.S. Martsenyuk, Phase diagrams of the binary systems of rare earth metals with tin, Fazovye Ravnovesiya, Strukt. Svois*tva Splav.* (1990) 70–100.
[35] V.N. Eremenko, M.V. Bulanova, P.S. Martsenjuk, Phase diagram of the terbium-tin system, Dokl. Akad. Nuak Ukr. SSR, B 1 (1990) 35–37.
[36] A. Palenzona, P. Manfrinetti, The Tin-rich side of the rare earth-tin systems (R = Gd, Tb, Dy, Ho, Er, Tm, Lu and Y), J. Alloy. Compd. 201 (1–2) (1993) 43–47.
[37] H. Okamoto, Comment on Sn-Tb (Tin-Terbium), J. Phase Equilib. 16 (1) (1995) 102–103.
[38] A. Szytula, D. Kaczorowski, L. Gondek, et al., Magnetic properties of RSn2 (R = Tb, Dy) compounds, Acta Phys. Pol. A 117 (4) (2010) 586–589.
[39] S.V. Meschel, O.J. Kleppa, Standard enthalpies of formation of some carbides, silicides, germanides, stannides and borides of terbium determined by high temperature direct synthesis calorimetry, J. Alloy. Compd. 234 (1996) 137–142.
[40] V.N. Eremenko, M.V. Bulanova, P.S. Martsenjuk, The dysprosium-tin phase diagram, J. Alloy. Compd. 189 (2) (1992) 229–233.
[41] V.V. Romaka, A. Tkachuk, V. Davydov, Interaction of the components in the Dy–Ag–Sn ternary system at 870K, J. Alloy. Compd. 439 (2007) 128–131.
[42] V.V. Romaka, P. Rogl, L. Romaka, et al., Interaction of the components in Dy–Ni–Sn ternary system and crystal structure of new compounds, J. Alloy. Compd. 485 (2009) 275–279.
[43] M. Bulanova, Y. Fartushna, K. Meleshevich, et al., Isothermal sections at 1400, 1100 and 900 °C of the Ti–Dy–Sn system below 40 at% Sn, J. Alloy. Compd. 578 (2013) 261–266.
[44] O.I. Bodak, V.V. Romaka, A.V. Tkachuk, et al., Phase equilibria in the Dy–Cu–Sn ternary system, J. Alloy. Compd. 395 (2005) 113–116.
[45] L.C.J. Pereira, D.P. Rojas, J.C. Waerenborgh, Isothermal section of the Dy–Fe–Sn phase diagram at 800 °C, J. Alloy. Compd. 396 (2005) 108–113.
[46] Y.H. Zhuang, J.M. Zhu, J.L. Yan, et al., Phase relationships in the Dy–Co–Sn system at 773 K, J. Alloy. Compd. 459 (2008) 461–465.